Chapter 8 • Logic

Solved Exercises

MCQs

1. Which expression is often related to inductive reasoning?
a) based on repeated experiments
b) if-and-only-if statements
c) a statement proven by a theorem
d) based on general principles
Ans: a
Induction generalizes from repeated observations or experiments.

2. Which sentence describes deductive reasoning?
a) general conclusions from limited observations
b) based on repeated experiments
c) based only on accurate units of information
d) draw conclusions from well-known facts
Ans: d
Deduction proceeds from accepted general facts to a conclusion.

3. Which statement is true?
a) The set of integers is finite
b) Interior angles of every quadrilateral sum to 180°
c) $7 \in Q$
d) All isosceles triangles are equilateral
Ans: c
Every integer is a rational number; the other claims are false.

4. The best negation of "The stove is burning" is:
a) The stove is not burning
b) The stove is dim
c) The stove is on low heat
d) It is both burning and not burning
Ans: a
Negation denies the original statement without adding a new condition.

5. The conjunction of p and q is true when:
a) both are false
b) both are true
c) only q is true
d) only p is true
Ans: b
$p \wedge q$ requires both components to be true.

6. A conditional is false only when:
a) antecedent true and consequent false
b) consequent true and antecedent false
c) antecedent true only
d) consequent false only
Ans: a
This is the defining truth-table rule for implication.

7. The contrapositive of $q \rightarrow p$ is:
a) $q \rightarrow \sim p$
b) $\sim q \rightarrow p$
c) $\sim p \rightarrow \sim q$
d) $\sim q \rightarrow \sim p$
Ans: c
Swap and negate both components.

8. "Every integer greater than 2 is a sum of two prime numbers" is a:
a) theorem
b) conjecture
c) axiom
d) postulate
Ans: b
This Goldbach-type assertion is presented as an unproved conjecture.

9. "A straight line can be drawn between any two points" is a:
a) theorem
b) conjecture
c) axiom

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d) logic
Ans: c
It is accepted as a basic geometric postulate.

10. "The sum of the interior angles of a triangle is 180°" is a:
a) converse
b) theorem
c) axiom
d) conditional
Ans: b
It has a deductive geometric proof.

2. Write the converse, inverse and contrapositive of each conditional.
(i) ~p
⇒ q: converse q
⇒ ~p
inverse p
⇒ ~q
contrapositive ~q
⇒ p.
(ii) q
⇒ p: converse p
⇒ q
inverse ~q
⇒ ~p
contrapositive ~p
⇒ ~q.
(iii) ~p
⇒ ~q: converse ~q
⇒ ~p
inverse p
⇒ q
contrapositive q
⇒ p.
(iv) ~q
⇒ ~p: converse ~p
⇒ ~q
inverse q
⇒ p
contrapositive p
⇒ q.

3. Write truth tables for (i) ~(p∨q)∨~q, (ii) ~(~q∨~p), and (iii) (p∨q)↔(p∧q).
Evaluate row by row in the order TT, TF, FT, FF. The table below gives the final value of each complete expression.
Algebraic check: (i) simplifies to ~q
(ii) simplifies to p∧q
(iii) has the same values as p↔q.

p	q	(i) ~(p∨q)∨~q	(ii) ~(~q∨~p)	(iii) (p∨q)↔(p∧q)
T	T	F	T	T
T	F	T	F	F
F	T	F	F	F
F	F	T	F	T

4. Differentiate a mathematical statement from its proof, with two examples.
A mathematical statement is a declarative sentence that is true or false. A proof is the logical argument establishing its truth.
Example 1 — statement: 2+3=5; proof: accepted arithmetic addition evaluates the left side as 5.
Example 2 — statement: the angles of a triangle sum to 180°; proof: draw a parallel line through one vertex and use alternate angles and a straight angle.

5. Differentiate an axiom from a theorem, with examples.

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An axiom (postulate) is accepted as a starting assumption without proof within a chosen system.
A theorem is derived from axioms, definitions and previously proved results by logical reasoning.
Axiom example: exactly one straight line passes through two distinct points. Theorem example: Pythagoras' theorem.

6. Explain the importance of logical reasoning in mathematical proofs, with an example.

Logical reasoning ensures that every step follows from definitions, assumptions or earlier results, so examples are not mistaken for a general proof.
If $x=2k+1$ is odd, then $x^2=4k^2+4k+1=2(2k^2+2k)+1$, which is odd. This proves the claim for every odd integer, not just tested values.

7. Classify each statement as axiom, conjecture or theorem.

(i) Exactly one line through two distinct points: axiom/postulate, a starting geometric assumption
(ii) Every even number greater than 2 is a sum of two primes: Goldbach conjecture, not a generally proved theorem
(iii) Triangle angle sum 180°: theorem, proved deductively in Euclidean geometry.

8. Prove the three algebraic identities deductively.

(i) $(x-4)^2+9 = x^2-8x+16+9$
$= x^2-8x+25.$
(ii) $(x+1)^2-(x-1)^2 = (x^2+2x+1)-(x^2-2x+1)$
$= 4x.$
(iii) $(x+5)^2-(x-5)^2 = (x^2+10x+25)-(x^2-10x+25)$
$= 20x.$

9. Prove each rational identity, justifying cancellations.

(i) $\frac{4+16x}{4} = \frac{4(1+4x)}{4}$
$= 1+4x.$
(ii) $\frac{6x^2+18x}{3x^2-27}=6x(x+3)/[3(x-3)(x+3)]=2x/(x-3)$, for $x\neq\pm3$.
(iii) $\frac{x^2+7x+10}{x^2-3x-10}=(x+2)(x+5)/[(x-5)(x+2)]=(x+5)/(x-5)$, for $x\neq-2,5$.

10. If integer x is odd, prove that 9x+4 is odd.

Let $x=2k+1$. Then $9x+4=18k+9+4=18k+13=2(9k+6)+1$.
9x+4 is odd.

11. If integer x is odd, prove that 7x+5 is even.

Let $x=2k+1$. Then $7x+5=14k+7+5=14k+12=2(7k+6)$.
7x+5 is even.

12. Prove the parity statements.

(i) If x is odd, x^2 is odd, while $-4x$ and 6 are even; odd+even+even is odd. Thus x^2-4x+6 is odd.
(ii) If $x=2k$, then $x^2+2x+4=4k^2+4k+4=2(2k^2+2k+2)$, an even integer.
even.

13. For non-empty sets A and B, prove $(A\cap B)'=A'\cup B'$.

$x\in(A\cap B)' \Leftrightarrow x\notin A\cap B \Leftrightarrow$ it is false that $x\in A$ and $x\in B$.
Thus $x\notin A$ or $x\notin B \Leftrightarrow x\in A'$ or $x\in B' \Leftrightarrow x\in A'\cup B'$.

14. If x,y are positive real numbers and $x^2<y^2$, prove $x<y$.

$y^2-x^2>0 \Rightarrow (y-x)(y+x)>0$. Since $y+x>0$, we must have $y-x>0$.
x<y.

15. Prove that the interior angles of a triangle sum to 180°.

Through vertex A of triangle ABC, draw a line parallel to BC.
The angles formed at A on the two sides of AB and AC equal $\angle B$ and $\angle C$ respectively by alternate interior angles.
Together with $\angle A$, they form a straight angle. Hence $\angle A+\angle B+\angle C=180°$.
Triangle angle sum=180°.

16. For nonzero real numbers, prove the fraction identities.

(i) $a/b=c/d \Leftrightarrow ad=bc$, multiplying both sides by bd; this also requires $d\neq0$.

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(ii) (a/b)(c/d)=ac/bd by multiplication of numerators and denominators.
(iii) a/b+c/b=(a+c)/b because the fractions have a common nonzero denominator.

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